$m_u = 1794 \text{ kNm}$

A post tensioned bridge girder with unbonded tendons is of box section of overall dimensions 1200 mm wide by 1800 mm deep with wall thickness of 150 mm. The high tensile steel has an area of 4000 mm^2 and is located at an effective depth of 1600 mm. The effective prestress in steel after all losses is 1000 N/mm² and the effective span of the girder is 24 m. If $f_{ck} = 40 \text{ N/mm}^2$ and $f_p = 1600 \text{ N/mm}^2$

estimate the ultimate flexural strength of the section.

Given data:

$f_{pe} = 1000 N/mm^2$

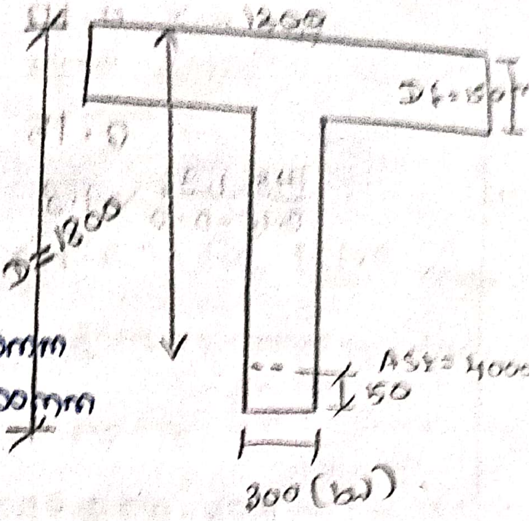
$l = 24 m$

$f_{ck} = 40 N/mm^2$ $b = 1200 mm$

$b_w = 300 mm$

$f_p = 1600 N/mm^2$

$A_p = 4000 mm^2$



$$\frac{\text{effective span}}{d} = \frac{24000}{1600} = 15$$

formula:-

$$M_u = f_{pu} A_{pw} (d - 0.42 x_u) + 0.45 f_{ck} (b - b_w) D_f (d - 0.5 D_f)$$

To find $A_p = A_{pw} + A_{pf}$

$A_{pw} = (A_p - A_{pf})$

$$A_{pf} = 0.45 f_{ck} (b - b_w) D_f / f_p$$

0.45 × 40

18 × 900

$$= 0.45 \times 40 (1200 - 300) 150 / 1600$$

$$A_{pf} = 1518.75 mm^2$$

$$A_{pw} = A_p - A_{pf}$$

$$= 4000 - 1518.75$$

$$A_{pw} = 2481.25 mm^2$$

To find effective reinforcement ratio

$$\frac{A_{pw} \cdot f_{pe}}{b_w \cdot d \cdot f_{ck}} = \frac{2481.25 \times 1000}{300 \times 1600 \times 40}$$

$= 0.129$

∴ unbonded tendons, from Table 7.2 from IS 1343

		$l/d = 10$	$l/d = 20$
for	0.10 →	1.45	1.26
find	0.129 →		l/d 15
	0.15 →	1.36	1.20

$$\frac{1.45-1.36}{0.15-0.10} = 1.8$$

$0.129 - 0.1 = 0.029$

$0.029 \times 1.8 = 0.0522$

$1.45 - 0.05 = 1.4$

$1.26 - 0.05 = 1.21$

$x_u/d = 0.41$ $\qquad \dfrac{f_{pu}}{f_{pe}} = 1.3 \times 1000 = 1300\ N/mm^2$

0.10 → 0.30 $\quad$ 0.36

0.129 → 0.40 $\quad$ 0.442

0.15 → 0.46 $\quad$ 0.52

$0.46 - 0.52 / 0.05 = 28 \times 0.029 = 0.0812$

$x_u = 0.41 \times 1600 = 656\ mm$

To find mu

$mu = 1300 \times 34,841.25 (1600 - 0.42 \times 656)$

$+ 0.45 \times 40 (1200 - 300) 150 (1600 - 0.5 \times 150)$

$mu = 8011 \times 10^6\ Nmm$

$mu = 8011\ kNm$